

Quantifying Model, Label, and Methodological Contributions to Variance in Real-Time Sepsis Onset Prediction

Abdul Ahad

Masters Thesis · M.Sc. in Computing
Department of Computer Science & Applied Physics
Atlantic Technological University (ATU), Galway

I. Introduction: Accuracy Survives, Usefulness Does Not

- Sepsis kills about **eleven million people a year**, nearly one death in five worldwide [1]; in the ICU every hour of delayed treatment raises the risk of death [2].
- Wang et al. [3] reviewed **91** real-time sepsis prediction models:

Metric	Internal	External
Median AUROC	0.811	0.783
Median Utility Score	+0.381	−0.164

- A negative Utility Score means the alerts cause more harm than benefit; AUROC did not detect the collapse, because it measures ranking, not whether alerts are timely or useful.
- The same gap appears in deployment: Epic Sepsis Model v2 across 227,091 encounters, AUROC 0.82–0.92 with PPV 0.13–0.26 [4].

“Which model is best?” is the wrong question to lead with.

II. Background: Limitations of the Conventional Approach

Choose one Sepsis-3 definition → use one train/test split → report AUROC → compare models.

1. **The sepsis label is not fixed.** Different ways of applying Sepsis-3 to the same database gave 867–2,178 cases [5].
2. **The data split can inflate results.** Sepsis prediction suffers the largest drop of any clinical task under a temporal shift [6].
3. **AUROC hides the real failure.** A model can keep a high AUROC while its clinical usefulness becomes harmful [3].
4. **The label captures doctor behaviour.** Models may learn when doctors suspect sepsis and treat, not when the patient worsens [7].

This thesis

Instead of fixing the label, the split and the metric and asking which model wins, it treats all of them as **sources of variance** and estimates each contribution **with an interval attached**, so that the ordering between them is established rather than assumed.

III. Research Question and Pre-Registered Hypotheses

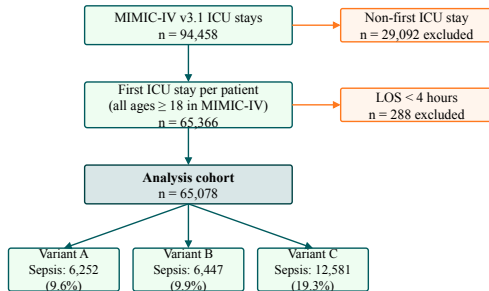
Research Question

In real-time sepsis prediction on MIMIC-IV [8], how much of the reported performance variation comes from (a) the choice of model, and (b) the researcher's decisions: how sepsis is defined, how data is split, when prediction is made, and how performance is measured?

- **H1 (Label dominance):** changing the sepsis definition affects AUROC at least as much as changing the model.
- **H2 (Coefficient instability):** at least one feature changes direction, or magnitude $\geq 2\times$, across label variants.
- **H3 (Treatment leakage):** restricting evaluation to before treatment reduces AUROC by ≥ 0.03 .
- **H4 (Split optimism):** temporal testing gives ≥ 0.02 lower AUROC than random testing.
- **H5 (Metric divergence):** Utility declines more than AUROC from internal to external validation.
- **H6 (Model-class null):** the primary statistical model is not outperformed by gradient-boosted trees by more than 0.02 AUROC.

One confirmatory family, Holm–Bonferroni ($m = 6$); H2 and H3 reserve slots without p-values, and H5's displayed contrast is a post-hoc substitute. **Scope:** the models estimate the hazard of onset in the *current* hour, i.e. detection at the recorded onset, not advance warning.

IV. Materials: Data and Cohort



- **Development:** MIMIC-IV v3.1 [8], a single centre (BIDMC, Boston).
- **Inclusion:** age ≥ 18 , first ICU stay per patient, length of stay ≥ 4 h $\Rightarrow$ 65,078 **stays**.
- **External:** eICU-CRD v2.0 [9], over 200 US ICUs, 181,589 unit stays; models applied frozen, without retraining.
- **Person-hour panel:** 72 h horizon; discharge and death are competing events, not censoring.
- **22 covariates** (6 vitals, 5 labs, 2 vasopressor flags, 3 six-hour slopes, 6 test-ordered indicators), 29 terms per outcome.

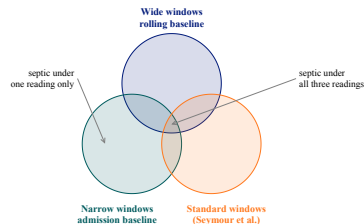
IV. Materials: Three Pre-Registered Sepsis-3 Label Variants

ID	Name	ABX→cx	cx→ABX	SOFA−	SOFA+	Baseline	Septic stays
A	Narrow	24 h	24 h	24 h	12 h	Admission	6,252 (9.6%)
B	Seymour-Std	72 h	24 h	48 h	24 h	Admission	6,447 (9.9%)
C	Liberal	72 h	72 h	48 h	24 h	Rolling 24 h	12,581 (19.3%)

ABX = antibiotic; cx = culture; SOFA−/SOFA+ = hours before/after suspicion time in which a SOFA rise of ≥ 2 must be observed.

- All three: **suspected infection** (antibiotic paired with a culture inside the windows) **and** a SOFA rise of ≥ 2 points [5], [10].
- **Variant B is the primary analysis**; A and C are pre-specified alternatives.
- Onset time is the suspicion time t_{susp} : the SOFA rise gates *membership*, not timing.

A and B have near-identical prevalence but agree on which stays are septic at only $\kappa = 0.267$.



V. Methods: Models and Validation Architecture

Primary model: dynamic discrete-time competing-risks hazard

Multinomial pooled logistic regression over person-hours (sepsis / discharge / death), natural-spline baseline in elapsed hours [11], [12]. The unpenalised fit is completely separated, so a small ridge penalty is required for identification.

Comparators: gradient-boosted trees (XGBoost, fixed conventional hyperparameters; boosting rounds chosen by early stopping on a 15% stay-level holdout within the *training* data), Cox proportional hazards, NEWS2, qSOFA and SIRS [13].

Split	How	Why
Internal (primary)	Time-based: train 2008–2016, test 2017–2019 [6]	Honest internal estimate
Internal (secondary)	Random 75/25 patient-level split (seed 42)	Split optimism (H4)
Treatment-anchored	Hours before first culture or antibiotic [7]	Treatment leakage (H3)
External	eICU-CRD v2.0, no retraining [9]	Generalisability (H5)

All splits are patient-level: no patient appears in both training and test sets.

V. Methods: Metrics and Statistical Inference

Primary metric: adapted PhysioNet-style Utility Score [14]

$$\tilde{U} = \frac{U - U_{\text{inact}}}{U_{\text{opt}} - U_{\text{inact}}} \quad \tilde{U} = 1 \text{ perfect, } \tilde{U} = 0 \text{ no-alert strategy, } \tilde{U} < 0 \text{ worse than no alerts}$$

- Rewards early true alerts in $[t_{\text{onset}} - 6, t_{\text{onset}}]$, penalises late and false ones.
- Reported at the 0.5 cut-off **and** at its maximum over a 60-point alert-rate grid, with the alert burden against the benchmark of 1.4 false alerts per true one [15].
- AUROC, AUPRC and calibration are reported alongside.

Inference and reporting

- Stay-level cluster bootstrap, $B = 2,000$; H1–H6 Holm-corrected ($m = 6$, FWER 0.05); exploratory families Benjamini–Hochberg.
- Pre-registered protocol; post-hoc protocol amendments are declared as such; TRIPOD+AI and PROBAST+AI reporting [16], [17].

VI. Results: Internal Temporal Performance (Variant B)

Model	AUROC	95% CI	Utility (best)	Alert burden at best
Primary	0.7590	[0.7434, 0.7749]	0.025	47.3
GBT	0.7424	[0.7265, 0.7583]	0.031	51.8
Cox	0.6444	[0.6271, 0.6614]	0.000	∞
NEWS2	0.6415	[0.6248, 0.6589]	0.000	∞
SIRS	0.6114	[0.5950, 0.6281]	-0.035	135.8
qSOFA	0.6091	[0.5934, 0.6254]	-0.252	248.8

Temporal test set 2017–2019: 548,827 person-hours, 1,040 onsets. Utility (best) = maximum over the threshold grid; ∞ = the best strategy is to raise no alert. Benchmark burden: 1.4 [15].

- The marginal intervals of the primary model and GBT overlap, but the paired contrast favours the primary model by 0.0165 AUROC (H6); both are separated from the four remaining models.
- Calibration is the primary model's strongest result: slope 1.055 under Variant B (ideal = 1).

VI. Results: The Variance Decomposition

Factor varied (others fixed)	Δ AUROC	95% CI	H
Model class (label B, temporal)	0.1499	[0.134, 0.171]	n/a
External transport (primary, label B)	0.0630	[−0.0215, 0.1563]	H5
Label definition (primary, temporal)	0.0123	[0.003, 0.032]	H1
Split method (primary, label B)	−0.0165	[−0.0372, 0.0040]	H4
GBT — primary (label B, temporal)	−0.0165	[−0.0308, −0.0012]	H6

Ranges, not variance components: they cannot be summed or expressed as shares of a total. The model-class range spans six heterogeneous specifications, the label range three Sepsis-3 readings; even so, the narrowest contrast across the gap (GBT vs Cox, 0.7424 vs 0.6444) exceeds the whole label range.

- **Model class dominates** across the six specifications compared.
- Transport ranks second on the point estimate, but its interval is the widest and covers zero: its rank is *not* established.
- The two decisions this literature scrutinises most, **label reading and split design**, move **discrimination least**.

VI. Results: Corrected Confirmatory Verdicts

H	Contrast	Est.	95% CI	p_{Holm}	Verdict
H1	Label spread — model spread	−0.1375	[−0.1623, −0.1083]	1.000	NOT confirmed
H2	Unstable covariates (count)	6	n/a	n/a	Descriptive
H3	Pre-treatment onsets (count)	0	n/a	n/a	Definitional
H4	Random — temporal AUROC	−0.0165	[−0.0372, 0.0040]	1.000	NOT confirmed
H5 [†]	Internal — external AUROC	0.0630	[−0.0215, 0.1563]	0.551	Not evaluable as pre-registered; substituted AUROC contrast inconclusive
H6	GBT — primary AUROC	−0.0165	[−0.0308, −0.0012]	1.000	H6 not rejected

[†] Post-hoc substituted contrast occupying H5's slot: the proportional utility-versus-AUROC contrast cannot be formed, because there is no internal Utility Score to decline from.

- **H1 reversed:** model spread 0.1499 vs label spread 0.0123, more than tenfold; the interval on the difference excludes zero.
- **H4 reversed:** the random split scores *lower* (0.7424 vs 0.7590); every model shows the same sign.
- **H6 not rejected:** primary 0.7590 vs GBT 0.7424. The upper limit lies inside the 0.02 margin, so one-sided **non-inferiority**, not two-sided equivalence.

VI. Results: H3, the Empty Pre-Treatment Window

Status: (NOT EVALUABLE AS PRE-REGISTERED), because its contrast cannot be formed.

Proposition (proved in the thesis)

Under variants A, B, C (and E) the onset is t_{susp} , the *later* element of an antibiotic–culture pair, so $t_{\text{onset}} \geq t_{\text{anchor}}$ for every labelled stay. The pre-treatment window therefore contains no onset **by construction**: a property of the implementation, not a measurement.

Var	Label	Pre-treatment person-h	Onsets	AUROC (primary / GBT)
B	Seymour-Std	179,003	0	undefined
F	Sepsis-3 Early-Onset	178,303	71	0.8081 / 0.8018
D	Deterioration	147,299	1,362	0.8498 / 0.9360

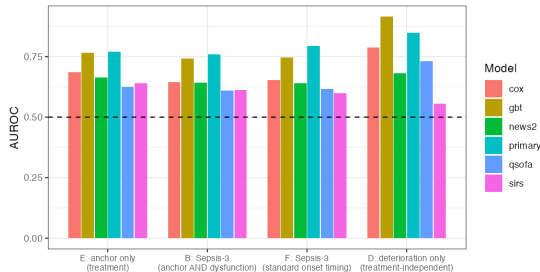
- **Variant F** (post-hoc): B's membership exactly, onset at the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$. 1,738 of 6,447 onsets move earlier (27.0%, median -14.0 h).
- Kamran et al. [7]: **confirmed for label membership, narrowed for label timing**. These AUROCs show the onsets exist and are separable, not that they are predictable in advance.

VI. Results: Splitting the Sepsis-3 Conjunction

Labels E, D and F are declared protocol amendments: (EXPLORATORY, POST-HOC)

Splitting the Sepsis-3 conjunction into its two limbs

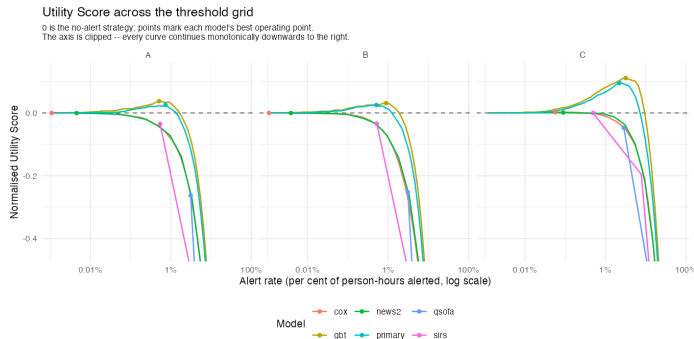
Temporal test set, unrestricted window; dashed line = chance



- **E (Anchor-Only)**, no SOFA gate, matches B's onset hour in 100.0% of B's stays: SOFA moves no onset.
- Anchor-attributable fraction:
 $\phi_{\text{anchor}} = 1.041 [0.990, 1.097]$ (primary, includes 1); $1.099 [1.037, 1.164]$ (GBT).
- **D (Deterioration)**, no treatment timestamp:
 $\kappa = 0.057$ with B. Its AUROC level is partly self-fulfilling, not a sepsis result.
- **F (standard onset timing)**: primary 0.7951 vs 0.7590; **not reproduced by GBT** ($\Delta = 0.0033$, $q = 0.559$).

For the primary model, onset *timing* matters more than the conjunction itself; for GBT the timing effect is not detected.

VI. Results: The Utility Ceiling



- At the 0.5 cut-off the primary model scores 0.000: it never alerts there, a threshold artefact.
- Swept over the grid, **no model exceeds 0.031 under label B** (0.111 under any variant), at 51.8 **false alerts per true one** vs a benchmark of 1.4 [15].
- Test-set thresholds: upper bounds, not deployed performance.

VI. Results: External Validation on eICU-CRD

Var	Internal (primary)	eICU primary	eICU GBT	eICU NEWS2	Events
A	0.7713	0.6913	0.7106	0.5866	27
B	0.7590	0.6960	0.6815	0.5259	29
C	0.7695	0.7370	0.6877	0.5074	73

Sepsis-3 arm on the microbiology-covered sub-cohort: frozen models, no retraining.

- **Label observability is the binding limitation:** culture documentation restricts the Sepsis-3 arm to 1.55% of eICU stays, leaving 29 Variant B events, below the 100-event floor.
- H5 substitute: degradation 0.0630 $[-0.0215, 0.1563]$, directionally consistent but inconclusive. Its causes are not decomposed.
- Sensitivity arms (different targets, never pooled): antibiotic-only (6,140 events) and Variant D (52,535 events) both degrade in the same direction.

VI. Results: Coefficient Instability and Equity

H2: coefficient instability (descriptive)

- 6 of 22 covariates unstable across labels: 2 by sign, 5 by magnitude.
- **Only 3 exceed their own estimation noise, all test-ordered indicators** [18]; the raw count is a loose upper bound.
- **Ablation** (post-hoc): dropping the 8 action-derived covariates costs -0.0076 AUROC (primary, B), while instability falls to 1 of 14. The cost is real but small.

Equity by label (exploratory)

- Subgroup AUROC: 6 contrasts clear the 100-event floor and **none survives** BH correction.
- Only two racial strata clear the floor, and one is *Unknown*: no powered comparison between two recorded demographic answers.
- Label sensitivity *does* survive: Unknown $+3.01$ pp ($q = 0.006$), Unable to obtain $+3.49$ pp ($q = 0.009$).
- Both are missingness categories, so the finding concerns **documentation completeness**, not race [19].

VII. Discussion: The Central Lesson and Limitations

The Sepsis-3 label on MIMIC-IV:

1. **Depends on clinical action for membership.** The empty pre-treatment window belongs to the onset rule: standard timing moves 1,738 onsets earlier, 71 of them into the test set's pre-treatment window (onset-hour separability, not advance warning).
2. **Is sensitive to how it is defined.** Cases nearly double across readings ($\kappa = 0.267$), yet AUROC moves by only 0.0123.
3. **Produces unusable alert behaviour.** No operating point comes near the 1.4 benchmark; the best under B costs 51.8 false alerts per true one.

These problems are independent of model complexity. Addressing them requires changing the label, not only the model.

Key limitations

- **Current-hour estimand:** detection at the recorded onset, not advance warning.
- **External validation is bounded by label transport:** 1.55% culture coverage, 29 events under B.
- **Single-centre development**, with a selected base cohort (first stays, ≥ 4 h).
- **The pre-registered onset rule is not the standard one;** F restores it post hoc.
- **Variant D is not a validated construct;** its components overlap the model inputs.
- **Post-hoc protocol amendments** are disclosed as such; the literature search is narrative, not systematic.

VIII. Conclusion: Contributions and Future Work

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- C1 Label and model-class spreads on the same test rows, each with an interval: the label moves membership ($\kappa = 0.267$), not discrimination (0.0123).
 - C2 Proof that the conjunction onset rule empties the pre-treatment window; standard timing places 71 test-set onsets there.
 - C3 Threshold-swept utility ceiling (0.031 under B) at 51.8 false alerts per true one.
 - C4 6 of 22 covariates unstable across labels; 3 survive their own noise, all order indicators.
 - C5 Equity-by-label on the same strata: the finding concerns documentation completeness, not race.
 - C6 Pre-registered hypothesis set; seed-fixed, reproducible pipeline with disclosed post-hoc amendments; TRIPOD+AI reporting; numbers are generated constants.
 - C7 Definitional vs. empirical label variance separated ($\kappa_{D,B} = 0.057$; $\phi_{\text{anchor}} = 1.041$).
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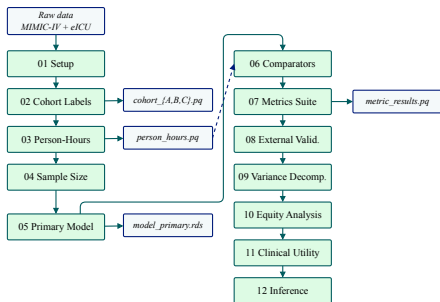
Future work: a sepsis definition grounded in objective physiology and adjudicated independently of the model's features; SEP-1 and CDC Adult Sepsis Event; prospective, protocol-blinded and multi-centre validation; utility-optimised training.

The premise inverted; the motivating problem did not. The label's consequence runs through *who* is counted and *when*, not through discrimination.

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Appendix: The Twelve-Stage Pipeline



- One orchestration script runs stages 01–12; one configuration module holds every path, identifier and threshold.
- Every reported number is emitted as a generated constant, so the text and the code cannot diverge.
- Frozen run: frozen-2026-08-31.
- Code and reproducibility:
github.com/ahadarif007/sepsis-variance-decomposition

Thank You

Questions & Discussion

Abdul Ahad

M.Sc. in Computing · Atlantic Technological University (ATU), Galway

Pre-registered hypothesis set · disclosed post-hoc amendments

Code: `github.com/ahadarif007/sepsis-variance-decomposition`